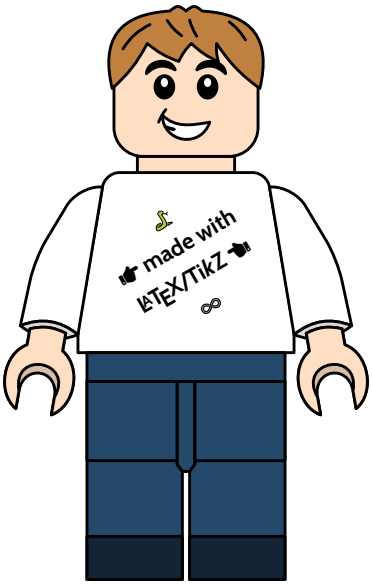
tikzbrickfigurines

Drawing brick-figurines with Ti*k*Z.

Version 0.1.1 – 18/08/2025

Cédric Pierquet
cpierquet – at – outlook . fr
https://github.com/cpierquet/latex-packages/tikzbrickfigurines



1 Introduction

The package allows to draw *brick*-figurines with Ti*k*Z.
The user can modify their colour, shape and elements.

The package is included in both TEXLive and MiKTEX and available from CTAN (https://ctan.org/pkg/tikzbrickfigurines).

The development version of this package is located on github at https://github.com/cpierquet/latex-packages/tikzbrickfigurines. If you have any problems, ideas or other feedback, please make constructive use of its bug tracker.

The style of this documentation is inspired by samcarter's doc for their great Ti*k*Zbricks package (https://ctan.org/pkg/tikzbricks).

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- Inspired by LEGO™, but all graphics are made from scratch.
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2 History

```
0.1.1: Enhancements for adjusted logo
0.1.0: Initial version
```

3 Loading, developments

3.1 Loading, useful packages

To use tikzbrickfigurines, simply use `\usepackage{tikzbrickfigurines}`.

```
\usepackage{tikzbrickfigurines}
```

Loaded packages are `tikz`, `simplekv`, `xstring`, `ifthen`, `listofitems`, `graphicx`, `etoolbox` and `calc`. Loaded libraries are `shapes.geometric,bbox,decorations,decorations.pathmorphing,calc`.

3.2 Developments

For the moment, the package uses internal macros for creating/manipulating pieces and elements, but eventually, the future plan is:

- to manipulate these elements with a `pic` code;
- to improve the code for the logo (work in progress).

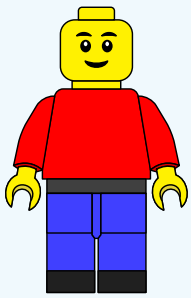
4 Global usage

4.1 Version 1

An individual Ti*k*Zbrickfigurines can be drawn with:

```
\tikzbrickfigurine%
  [keys]%
  <tikz options>%
  {height}
```

```
%default output
\tikzbrickfigurine{1.5in}
```

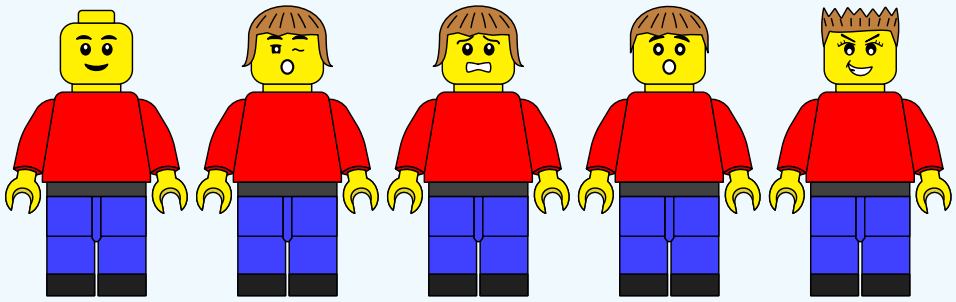


4.2 Version 2

An individual Ti*k*Zbrickfigurines can be drawn with:

```
\minitikzbrickfigurine%
  [keys]%
  <tikz options>%
  {height}%
  {list of styles, within M/E/Eb/H}
```

```
\minitikzbrickfigurine{1.5in}{1/1/1/0}%default style
\minitikzbrickfigurine{1.5in}{4/7/1/2}%with choosen styles
\minitikzbrickfigurine{1.5in}{3/*/9/*}%with choosen and random styles
\minitikzbrickfigurine{1.5in}{*}%with full random styles
\minitikzbrickfigurine{1.5in}{*}%with other full random styles
```



4.3 Individual part

An individual Ti*k*Zbrickfigurines piece can be drawn with:

```
\tikzbrickfigurinepiece%
  [keys]%
  <tikz options>%
  {piece / style}%
  {height}
```

```
\tikzbrickfigurinepiece[head=true]{hair=3}{1in}%
\tikzbrickfigurinepiece[head=true]{eyes=7}{1in}%
\tikzbrickfigurinepiece[head=true]{mouth=2}{1in}%
\tikzbrickfigurinepiece{head=2/3/*/5}{1in}%
```

5 Keys and styles

5.1 Keys

Available **keys** are:

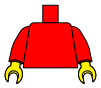
- `color lines`: eventual color for borders
- `color skin`: global color for skin (override individual colors if given)
- `color sweat`: global color for body and arms (override individual colors if given)
- `color pant`: global color for pant (override individual colors if given)

- `color belt`: color for the belt (`darkgray`)
- `color bottom belt`: color for the belt (`blue!75`)
- `color body`: simple color for the body (`red`)
- `color arm`: simple color for the arms (`red`)
- `color arm shadow`: simple color for the arms shadow (`red!85!black`)
- `color hand`: simple color for the hand (`yellow`)
- `color hand shadow`: simple color for the hand shadow (`yellow!85!black`)
- `color neck`: simple color for the neck (`yellow`)
- `color head`: simple color for the head (`yellow`)
- `color hair`: color for the hair (`brown`)
- `color top pant`: simple color for top pant (`blue!75`)
- `color middle pant`: simple color for mid pant (`blue!75`)
- `color foot`: simple color for shoes (`darkgray!50!black`)
- `adjusted logo`: the logo will be adjusted to sweat
- `display logo`: boolean for logo (`false`)
- `custom logo`: boolean for custom logo (`false`)
- `logo`: code for the optional logo
- `scale logo`: custom scale for `display logo` (`0.667`)
- `empty`: boolean for empty figurine (`false`)
- `hairstyle`: style of hair, from 0 to 10 (`0`)
- `eyestyle`: style of eyes, from 1 to 10(`1`)
- `eyebrowstyle`: style of eyebrows, from 1 to 10(`1`)
- `mouthstyle`: style of mouth, from 1 to 10(`1`)
- `seed`: random seed for some hair code (`365479`)
- `bbox` boolean for using `bbox` library (`false`)
- `vcenter` boolean for centering vertically, with `tikz baseline` (`false`)
- `head`: boolean for showing head with individual element (`false`)

5.2 Elements

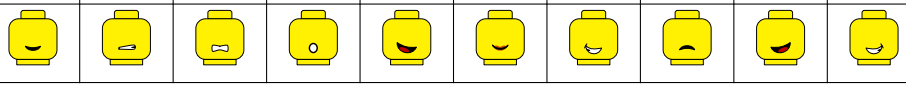
5.2.1 Body / legs / head / arms / hands / heads

```
\tikzbrickfigurinepiece[keys]<tizk options>{piece}{height}
```

piece	body	legs	head	left arm	right arm	left arm wo hand	right arm wo hand	left hand	righ hand
									

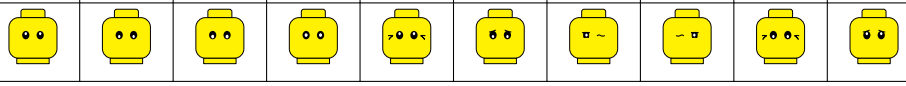
5.2.2 Mouth

```
\tikzbrickfigurinepiece[keys]<tizk options>{mouth=...}{height}
```

mouth=	1	2	3	4	5	6	7	8	9	10
										

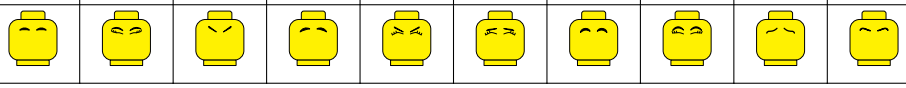
5.2.3 Eyes

```
\tikzbrickfigurinepiece[keys,head=true]<tizk options>{eyes=...}{height}
```

eyes=	1	2	3	4	5	6	7	8	9	10
										

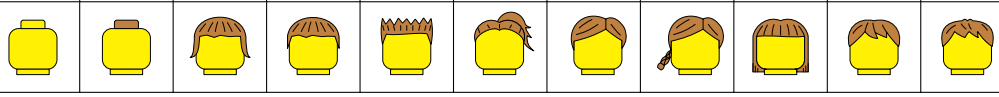
5.2.4 Eyebrows

```
\tikzbrickfigurinepiece[keys,head=true]<tizk options>{eyebrows=...}{height}
```

eyebrows=	1	2	3	4	5	6	7	8	9	10
										

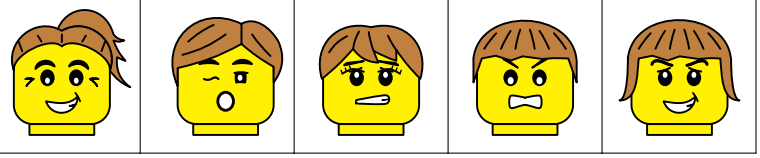
5.2.5 Hair

```
\tikzbrickfigurinepiece[keys,head=true]<tizk options>{hair=...}{height}
```

hair=	0	1	2	3	4	5	6	7	8	9	10
											

5.2.6 Head(s)

```
\tikzbrickfigurinepiece[keys]<tizk options>{head=...}{height}
```

head=	*	4/8/1/6	2/*/*/9	3/3/3/3	10/10/10/2
					

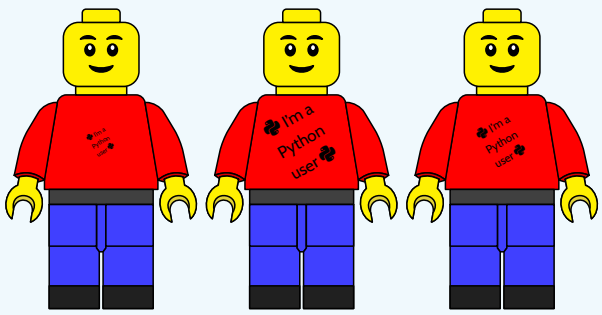
5.3 The optional logo

Due to scaling process, a global scale is applied to whole figure (with the Ti*k*Z option `transform shape`). So every node is scaled too.
The special key `scale logo=...` can adjust most precisely the logo!
There's three methods in order to place a logo:

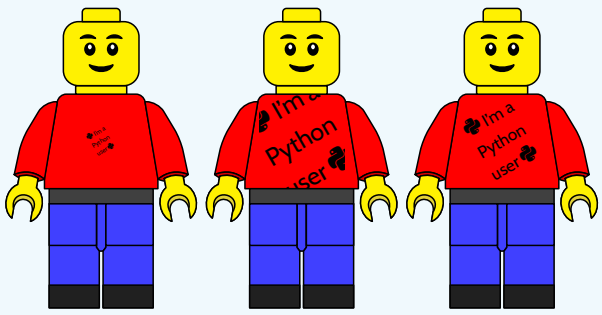
- boolean `custom logo` and key `logo=...` with the same global scale;
- boolean `display logo` and key `logo=...` with a *cancel* scale;
- key `adjusted logo=...` to adjust in the sweat.

```
\def\mylogo{\sffamily\rotatebox{30}{%
  \begin{tabular}{@{}c@{}}%
    \faPython\ I'm a \\ Python \\ user \faPython
  \end{tabular}}%
}

\mylogo
```

```
\tikzbrickfigurine[custom logo,logo=\mylogo]{4cm}%with 'global' scale
\tikzbrickfigurine[display logo,logo=\mylogo]{4cm}%with 'cancel' scale
\tikzbrickfigurine[adjusted logo=\mylogo]{4cm}%adjusted
```




```
\tikzbrickfigurine[custom logo,logo=\mylogo,scale logo=1]{4cm}%with 'global' scale
\tikzbrickfigurine[display logo,logo=\mylogo,scale logo=1]{4cm}%with 'cancel' scale
\tikzbrickfigurine[adjusted logo=\mylogo,scale logo=1]{4cm}%adjusted
```



6 Samples

6.1 Influence of options

```
\definecolor{brickskin}{HTML}{FFDFC4}

\def\mylogo{\rotatebox{30}{\sffamily\begin{tabular}{@{}c@{}}\faFastForward\ I'm a\\\LaTeX\\user
    \faFastBackward\end{tabular}}}

\tikzbrickfigurine%
  [%
    color lines=darkgray,
    color belt=lightgray,
    color bottom belt=blue!75,
    color body=red,
    color arm=violet,
    color arm shadow=lime,
    color hand=yellow,
    color hand shadow=orange,
    color neck=teal,
    color head=brickskin,
    color hair=cyan,
    color top pant=pink,
    color middle pant=pink!50!black,
    color foot=white,
    adjusted logo=\mylogo,
    scale logo=1,
    hairstyle=6,%
    eyestyle=5,%
    eyebrowstyle=10,%
    mouthstyle=2
  ]%
  <rotate=15>
  {2.75in}%
```

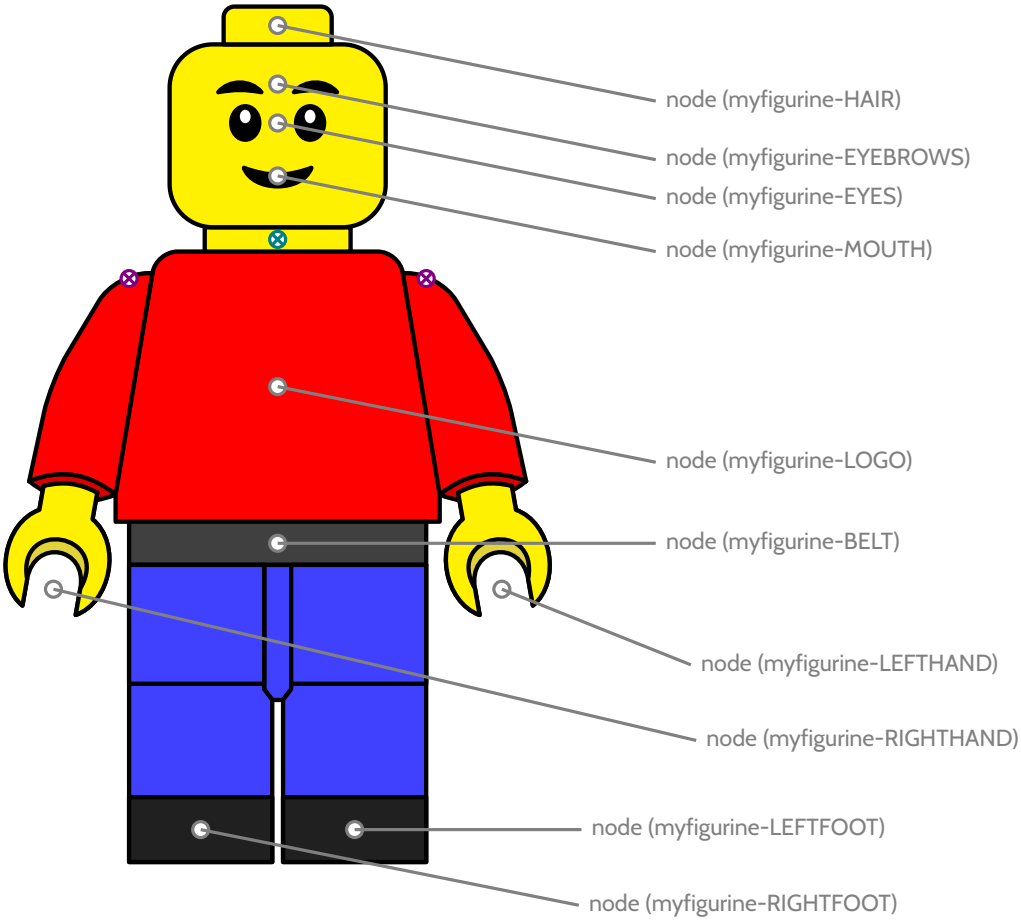
6.2 Random usage

```
\newcommand\randomcolor[1]{%
  \pgfmathsetmacro{\R}{random(0,10000)/10000}%
  \pgfmathsetmacro{\G}{random(0,10000)/10000}%
  \pgfmathsetmacro{\B}{random(0,10000)/10000}%
  \definecolor{#1}{rgb}{\R,\G,\B}%
}

\pgfmathdeclarerandomlist{listsksintones}%
  {{brickskin}{yellow}{pink}{brown}{brown!50!white}}
\pgfmathdeclarerandomlist{listhairtones}%
  {{lightgray!15}{yellow}{orange}{brown}{gray}{black}}
\pgfmathdeclarerandomlist{listlogos}%
  {{unicorn}{alien}{snake}{adhesive bandage}{desert island}{videocassette}{alien monster}}

%five 'random' clothes + face + skin + hair + logo brickfigurines
\foreach \i in {1,...,5}{%
  \randomcolor{randomsweatcolor}%
  \randomcolor{randompantcolor}%
  \randomcolor{randomshoecolor}%
  \pgfmathrandomitem{\skinrandom}{listsksintones}%
  \pgfmathrandomitem{\hairrandom}{listhairtones}%
  \pgfmathrandomitem{\logorandom}{listlogos}%
  \minitikzbrickfigurine%
    [%
      color skin=\skinrandom,%
      color hair=\hairrandom,%
      color sweat=randomsweatcolor,%
      color pant=randompantcolor,%
      color foot=randomshoecolor,%
      display logo,scale logo=0.8,%
      logo={\openmoji{\logorandom}},%
      vcenter
    ]%
    <rotate=\fpeval{randint(-20,20)}>%
    {1.75in}{*}
}
```

6.3 Overlay usage

Several nodes a created within internal code, and with TikZ options, overlays elements can be added.

```
\tikzbrickfigurine%
  [figure name=myfigurine]%
  <remember picture>
  {4.5in}%
```



Other available nodes are:

- `(<mynode>-RIGHTSHOULDER)`, `(<mynode>-LEFTSHOULDER)`;
- `(<mynode>-NECK)`.